AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
SPECIALIZED ENGINEERING PROGRAMS
COMMUNICATION ENGINEERING PROGRAM (COMM)

Major Task (Part2) **Total :10 Marks**

PHM312s: Complex variables, Special Functions and Partial Differential Equations

Fall2025 **1/20**

Name: **ID:** **Deadline: Week 14**

Please, Solve each problem in its assigned place ONLY (the empty space below it)

Problem Set No. 3
Functions of the complex variable

1. Find the real and imaginary parts of each of the following functions and state its domain

a) $f(z)=e^{z^2}$

b) $f(z)=2z^3-3z$

c) $f(z)=\dfrac{(z+i)}{(z^2+1)}$

2. Describe the image of the following regions under the specified function (Show the regions graphically)

a) $\mathrm{Re}(z)\geq 0$ for $f(z)=z+5$

b) $|\arg(z)|\leq \pi/4$, $|z|\leq 2$ for $f(z)=z^3$

6. Under the Reciprocal transformation, find the image of the following regions & show the regions (The pre-image & The image) graphically.
   a) The infinite strip $0 < y < 1/2c$

   b) The circle $|z-2|=1$

9. Using Cauchy – Riemann's equations, determine where the following functions are differentiable and where they are analytic? If the function is differentiable, find its derivatives.

a) $w = 2y - i\,x$

b) $w = e^{x}(\cos y + i \sin y)$

c) $w = z^{2}\,\bar{z}$

b) $\overline{f(z)}$ is analytic.

12. Show that each of the following functions is harmonic and find a corresponding analytics function $f(z) = u + i\,v$:

a) $u = 2\,x\,(1 - y)$

b) $u = \cos x\ \cosh y$

15. Find all the roots of the following equations:

a) $\cos z = 2$

b) $\cosh z = 0$

16. Find all the values of

a) $\ln(-1)$

b) $\ln(-e\,i)$

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Problem Set No. 4
Partial Differential Equations

Find the separated solutions to solve the initial – boundary value problem

$$u_{tt} = C^2 u_{xx} \qquad 0 \le x \le \pi\,, t \ge 0$$

Subject to the conditions: $u(0,t) = 0 \;\; u(\pi,t) = 0$

$$u(x,0) = 6\sin(3x)\,, u_t(x,0) = x^2 - \pi x$$

Consider Dirichlet problem in the unit square R for Laplace equation

$$u_{xx} + u_{yy} = 0, \;\; 0 < x < 1, \;\; 0 < y < 1$$

Boundary condition:

$u = 0$ *on the top of the square*

$u = 100$ *on the sides , and the bottom of the square*

Find the value of u *on the rest of unit square* R

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Problem Set No. 5
Bessel Functions

1. Solve in terms of Bessel functions the following differential equations:

a) $x^2\ y'' + x\ y' + (x^2 - 9)\, y = 0$

b) $x^2\ y'' + x\ y' + (x^2 - 8)\, y = 0$

c) $x^2\ y'' + x\ y' + (3x^2 - 4)\, y = 0$

d) $x^2\ y'' + x\ y' + 4(x^4 - n^2)\, y = 0 \quad , \quad n \in I$

e) $x\ y'' + 3\ y' + x\ y = 0$

c) $J_{3/2}(x) = \sqrt{\frac{2}{\pi x}}\left(\frac{\sin x}{x} - \cos x\right)$

d) $J_{-3/2}(x) = -\sqrt{\frac{2}{\pi x}}\left(\frac{\cos x}{x} + \sin x\right)$

3. Show that

a) $J_n''(x) = \frac{1}{4}\left(J_{n-2}(x) - 2\,J_n(x) + J_{n+2}(x)\right)$

5. Solve in terms of Bessel functions the following differential equations:

$$y'' + x\, y = 0$$